

Connor Savugot

B.S. in Computer Engineering | North Carolina State University | Graduated May 2026

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Computer engineering graduate with four internships and full-time engineering experience spanning embedded firmware, production support, and manufacturing-data automation. Develops C/C++ software, troubleshoots electrical and embedded systems, and builds tools and procedures for engineering and production users.

SKILLS

Programming & Data: C, C++, SQL, VBA, Microsoft Access, Excel, Python, MATLAB, Bash

Embedded Platforms: STM32, dsPIC33, PIC32, TI AM62, TMS320, MSP430; bare-metal firmware, FreeRTOS, Embedded Linux

Peripherals: GPIO, DAC control, ADC sensing and filtering, PWM control, timers, interrupts, watchdogs, and communications

Interfaces: CAN 2.0, CAN FD, DroneCAN, J1939, I2C, IPMI, SPI, UART/RS-232, Ethernet, TCP/IP

Hardware & Manufacturing: Schematic interpretation; hardware validation against datasheet specifications, PCB review in Altium, KiCad, and P-CAD; MCU programming/debug interfaces, oscilloscopes, logic/protocol analyzers; production programming, test automation, tooling data

EXPERIENCE

AEGIS Power Systems | Embedded Firmware Engineer

May 2026–Present

- ▷ Develop and debug C firmware; perform system testing using software logs, protocol analyzers, oscilloscopes, and multimeter measurements to validate hardware and software behavior.
- ▷ Isolate root causes and verify corrective actions by assessing system-level failures across PCBs, wiring, power, firmware, CAN/Ethernet, Linux services, and application software using schematics, datasheets, and diagnostic tools.
- ▷ Support firmware development, hardware integration, and documentation for VPX/IPMI power systems and TI AM62 Embedded Linux controllers; troubleshoot Yocto/Arago images, device trees, boot issues, CAN power devices, and watchdogs.

AEGIS Power Systems | Embedded Systems Intern | Computer Science Intern

Summers 2024 & 2025

- ▷ Wrote bare-metal C firmware for a TMS320-controlled bidirectional power converter with 12 PWM A/B pairs (24 outputs); configured frequency, period, duty cycle, and phase shift for each A output and set its paired B output to follow or invert it.
- ▷ Simplified TI's C++ CLI serial flasher for C2000 devices by removing unused device, CPU2, and dual-boot options, allowing production staff to program precompiled firmware with the click of a single button.
- ▷ Worked on a multi-processor power system where a FreeRTOS PIC32 managed communications and DSP firmware updates over TCP/IP, SPI, UART, and CAN/J1939, while a C2000 TI DSP executed real-time converter controls.
- ▷ Established Git-based development and release processes to maintain firmware-to-hardware traceability, preserve defect and revision history, support multi-engineer development, and provide production with controlled software releases and documentation.
- ▷ Developed a secure command-line environment for the SCB301 using Bash and Python on a stripped TI Yocto/Arago Linux system; created a chroot environment, restricted user access, product-management commands, and firmware-update functionality.

TEAM Industries | Engineering Intern

Summers 2021 & 2022

- ▷ Developed VBA-driven inspection tools that validated production measurements against engineering tolerances, enforced data-entry rules, and automatically classified inspected parts as ACCEPT, REJECT, or INCOMPLETE.
- ▷ Built a configurable validation and duplicate-detection system for an existing 17,000+ record plant-wide Tooling Where-Used List (TWUL), using advanced Excel formulas and conditional formatting to let users define comparison criteria and automatically identify, locate, and highlight duplicate or inconsistent records.

PROJECTS

EV Active Sensor Adapter | Battery Monitoring & CAN

NC State University

- ▷ Developed STM32G474 firmware for voltage and temperature sensing, LED indicators, state-machine logic, and error logging on a team-built EV sensor board; supported CAN integration.

C++ Maze Solver

NC State University

- ▷ Developed an object-oriented C++ maze-solving application using reusable classes, dynamic data structures, and multiple search strategies; compared correctness and iteration counts across implementations.

C++ Microarchitecture Simulators

NC State University

- ▷ Developed modular C++ simulators for processor subsystems using classes and objects to model system state, component interactions, and configurable behavior; validated results across multiple test configurations.

TEACHING EXPERIENCE

ECE 306 Embedded Systems | Teaching Assistant & Course Mentor

Spring 2025–Spring 2026

- ▷ Taught supplemental lessons and helped students debug MSP430FR2355 firmware and custom PCBs across three semesters, including ADC sensors, PWM motor control, timers, wiring, soldering, and toolchain problems.

IEEE LEADERSHIP

IEEE at NC State | OPS2 Lead & Parts Distribution Coordinator

Fall 2025–Spring 2026

- ▷ Led two semesters of OPS2, IEEE's ten-week applied electronics course that paired classroom theory with hands-on soldering, PCB and BOM development, embedded programming, PlatformIO, and GitHub.
- ▷ Built and managed an ECE 306 parts program that cut student costs and raised \$10,000 for IEEE in two weeks.